WORKSHOP 2: AUTOMATED DATA ANALYSIS & REPORTING

Automated Analysis of Life Expectancy and GDP per Capita in 2020

Author: Bertha Kembabazi
Date: 17 May 2026
Workshop provider: The Graph Courses

Purpose

This report was prepared for Workshop 2: Automated Data Analysis & Reporting, part of The Graph Courses Generative AI Workshop series. The purpose of the exercise was to use AI-assisted data analysis tools to turn country-level Gapminder data into a brief, readable report. Specifically, the report examines whether countries with higher GDP per capita also tend to have higher life expectancy in 2020.

Introduction

The two selected indicators are life expectancy at birth, measured in years, and GDP per capita, measured in international dollars. These indicators were selected because they represent two broad dimensions of development: population health and economic resources. Comparing them across countries helps illustrate how economic conditions are associated with health outcomes, while also showing that income alone does not fully explain differences in life expectancy.

Approach, Tools and Methodology

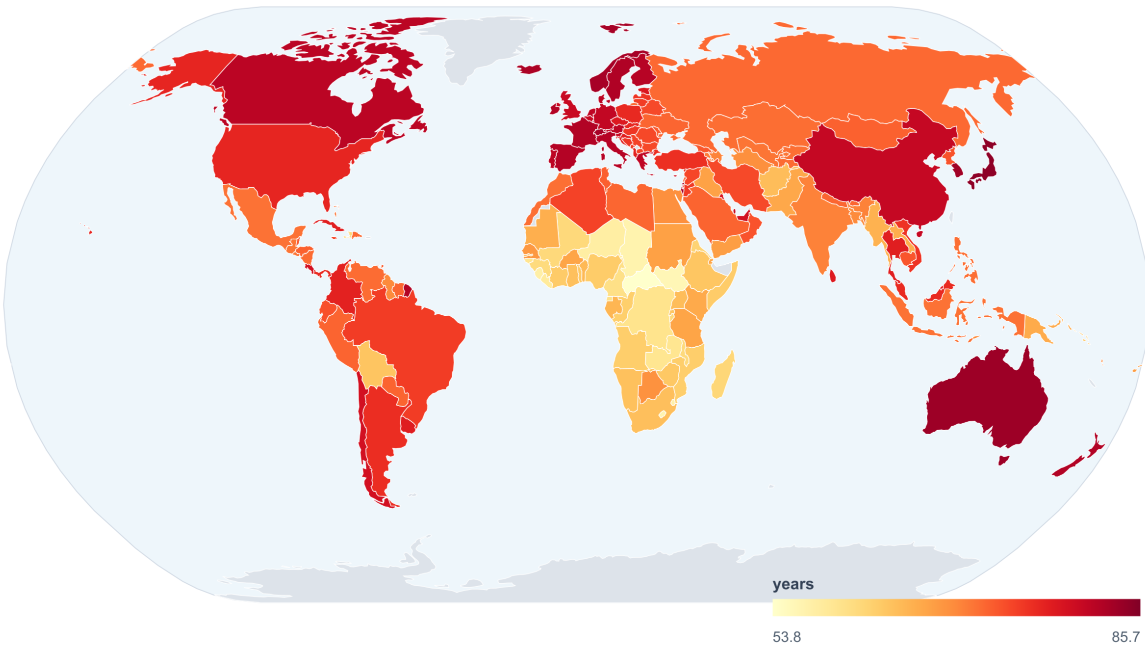
Step / Tool	How it was used
Data source	Gapminder country indicator CSV files: life expectancy and GDP per capita.
Data preparation	Viewed the raw country-year tables, then subsetted both indicators to 2020.
Validation	Checked that each 2020 subset had one row per country before mapping or merging.
Merge method	Joined the two 2020 datasets by Gapminder country code (geo), producing a matched analysis file.
Visualization tools	Used D3.js with world-atlas / Natural Earth shapes for actual world choropleth maps, and a scatterplot for the relationship.
Statistical approach	Calculated Pearson correlation using raw GDP, Pearson correlation using log10 GDP, and Spearman rank correlation.

Data check: the life expectancy file contained 194 countries, and the GDP per capita file contained 193 countries. For 2020, the life expectancy subset had 194 rows and the GDP subset had 193 rows. The merged scatterplot dataset contained 193 matched countries.

Figure 1. Life Expectancy World Map

Life expectancy at birth by country, 2020

Indicator 1: life expectancy at birth, in years



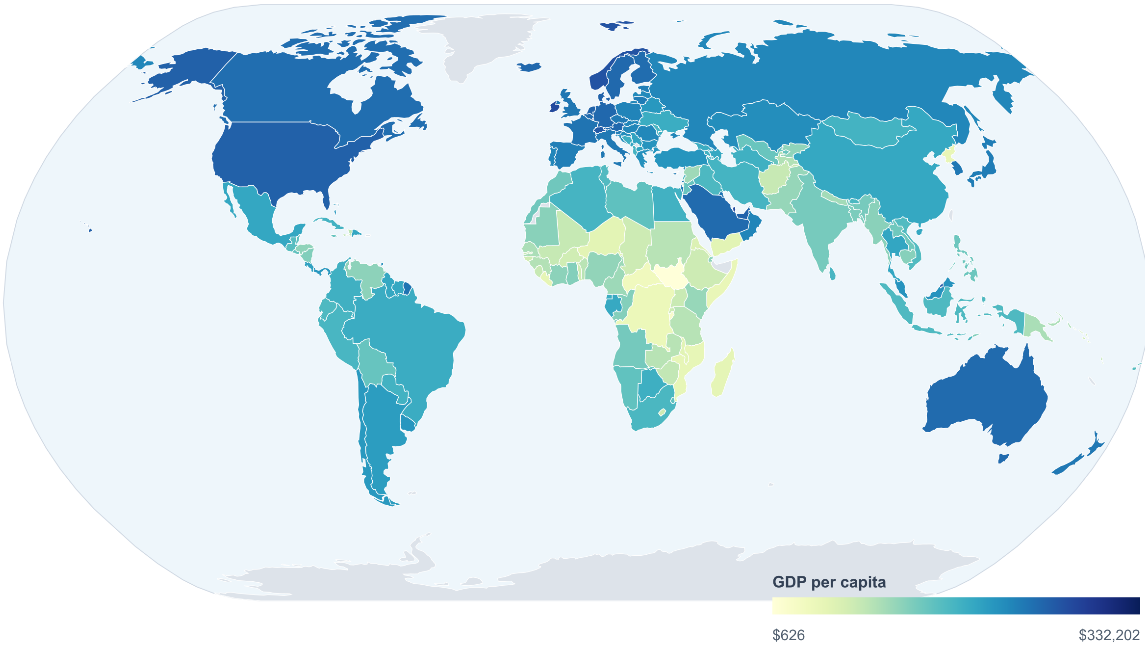
Actual country shapes from D3 world-atlas / Natural Earth. Gray areas indicate no matched 2020 data or territories outside the dataset.

Actual country-shape choropleth map of life expectancy at birth in 2020.

Figure 2. GDP per Capita World Map

GDP per capita by country, 2020

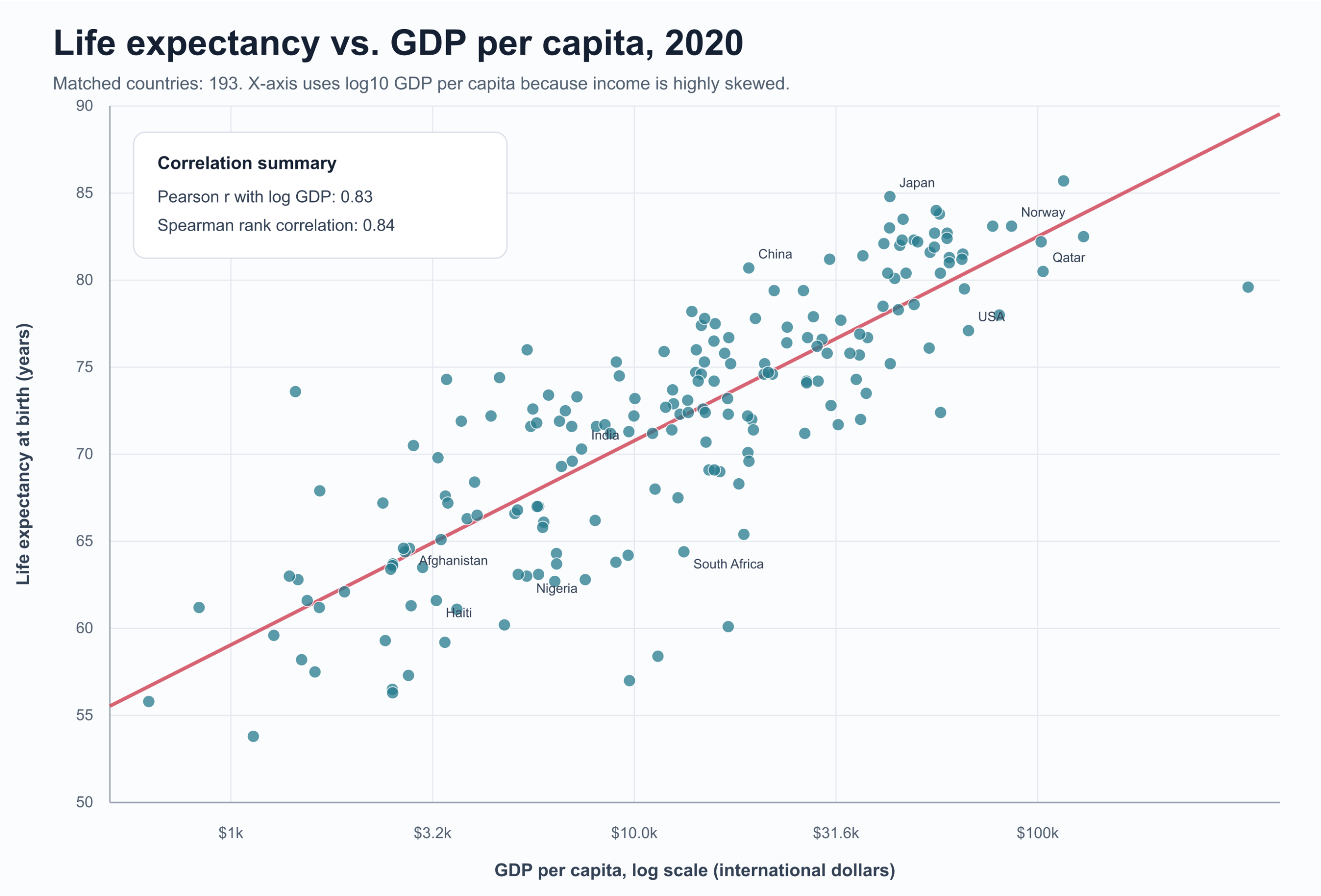
Indicator 2: GDP per capita, international dollars; log color scale



Actual country shapes from D3 world-atlas / Natural Earth. Gray areas indicate no matched 2020 data or territories outside the dataset.

Actual country-shape choropleth map of GDP per capita in 2020; color scale uses log GDP.

Figure 3. Relationship Between the Indicators



Scatterplot of life expectancy against GDP per capita in 2020, with GDP shown on a log scale.

Findings

• The maps show a broad global pattern: countries in Europe, North America, Australia/New Zealand, and parts of East Asia tend to have higher life expectancy and higher GDP per capita.

• Many countries in Sub-Saharan Africa have lower life expectancy and lower GDP per capita in 2020, although there is variation within the region.

• The scatterplot shows a strong positive relationship between GDP per capita and life expectancy. Countries with very low GDP per capita tend to have shorter average life expectancy.

• The relationship is stronger when GDP per capita is placed on a log scale. This suggests diminishing returns: increases in income appear to be associated with larger health gains at lower income levels than at already-high income levels.

Correlation Summary

Correlation measure	Value	Meaning
Pearson correlation using raw GDP per capita	0.58	Moderate positive relationship
Pearson correlation using log10 GDP per capita	0.83	Strong positive relationship
Spearman rank correlation	0.84	Strong positive ranked relationship

Interpretation

The 2020 data show a strong positive association between GDP per capita and life expectancy. The Pearson correlation using raw GDP per capita is moderate at 0.58, while the Pearson correlation using log10 GDP per capita is much stronger at 0.83. The Spearman rank correlation is also strong at 0.84. This means that countries with higher GDP per capita generally tend to have higher life expectancy, but the relationship is not perfectly linear. GDP is useful for understanding broad development patterns, yet it should not be treated as the only cause of longer lives. Health systems, inequality, education, conflict, public health policy, sanitation, and access to care also influence life expectancy.

Conclusion

Overall, the analysis supports the conclusion that economic development and health outcomes are closely linked. However, the flattening of the relationship at higher income levels shows that additional national income does not automatically translate into longer lives. Countries need both economic resources and effective systems that convert those resources into equitable health, education, and social protection outcomes.

Personal Reflection

Using AI tools made it easier to inspect the data, subset the indicators to 2020, calculate correlations, and generate an organized report structure. The most difficult part was quality control. I had to review the first visual outputs carefully and notice that the map needed to use actual country shapes rather than a simplified tile-style visualization. It was also important to check that the data were in the correct format before mapping and to use a log scale for GDP per capita because income values are highly skewed. What surprised me most was how much stronger the relationship became after using log GDP per capita. I learned that AI can speed up analysis and reporting, but the user still needs to guide the tool, ask for refinements, verify the figures, and make sure the interpretation is honest and does not overclaim causality.